

Ocean Acidification Lab Report

After completing part 1, answer the following questions.

1. What does pH stand for?
pH stands for the potential/power of hydrogen.
2. Explain what H^+ and OH^- have to do with the pH scale.
The pH scale determines how acidic (H^+) or basic (OH^-) the water is.
3. Does water absorb carbon dioxide? How do you know? Support your answer with your data.
Water absorbs carbon dioxide. I know this because water absorbs carbon dioxide by CO_2 bonding with the water to create carbonic acid, which dissolves into H^+ as well as HCO_3^- . My data supports this as the ocean water and the tap water changed color, indicating that it is more acidic because of the carbon dioxide.
4. Is carbon dioxide acidic or basic? How do you know? Support your answer with your data.
Carbon dioxide is acidic because when added to water in our experiment, it turned the water acidic. This implies that carbon dioxide itself is acidic.
5. Did the ocean water absorb CO_2 ? Support your answer with your data.
The ocean water absorbed the CO_2 because it changed into a lime color, which confirms the process of the ocean water absorbing carbon dioxide, and they bond to form carbonic acid. This is because the lime color change of the water indicates the water's acidity levels.
6. If you put the same amount of vinegar and baking soda into each system (tap and ocean), why weren't the results the same? Explain any suggestions you might have.

I hypothesize that the results weren't the same because ocean water contains a higher salt content than tap water does. Making the ocean water a higher acidity level than the tap water, as indicated by the lighter color difference.



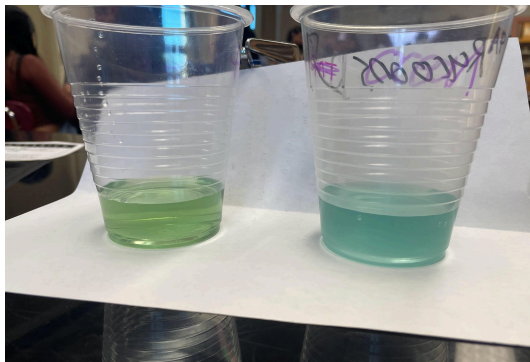
Left: Tap Water Before Vinegar
Right: Ocean Water Before Vinegar



Left: Tap Water After Vinegar
Right: Ocean Water After Vinegar

Carbon Dioxide	Tap Water	Ocean Water
pH levels Before Vinegar:	6	8
pH levels After Vinegar:	5	4

Part 2



Left: Tap Water Right Ocean Water

Water type	Time	Ph
Ocean	16.20 seconds	4
Tap	12.32 seconds	4

The addition of CO₂ caused the solutions to change colors due to the addition of the acidic pressure. The ocean system took 4 seconds longer to change color because of the carbonate buffer

system that ocean water has to maintain its pH levels, while tap water does not have that. The carbonate buffer system works when the ocean absorbs CO₂, and it bonds with the water to form carbonic acid, which can further break down into a hydrogen ion and a bicarbonate ion (this can go into reverse as well). Carbon dioxide stays in the ocean for a brief amount of time before dissolving, unlike tap water, where carbon dioxide has a much faster reaction. The ocean water began to turn back to green after a bit because of the carbonate buffer system that I had explained earlier; this system goes into effect to regulate its pH levels. The buffering system in the actual ocean comes from the reaction of extra hydrogen ions in the atmosphere. The pH increased to 5, making it less acidic. This means the buffer systems act as a way of preventing the ocean from becoming too acidic, but cannot completely remove the CO₂. This means that ocean animals are being affected by the acidity in the ocean because their bones contain calcium ions, which can bond with carbonate ions to create a compound, calcium carbonate. This is a normal process for animals to build up their bone structure, unless there are too many excess hydrogen ions that can lead to the disintegration of skeletal structures, which will lead to the killing of ocean life as a whole.